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🎓 EDUCATION

Technische Universität Nürnberg (UTN),

MSc in Artificial Intelligence and Robotics

10/2026 – Present

Charotar University of Science and Technology (CHARUSAT),

BTech in Information Technology (CGPA: 9.36 / 10.00)

06/2022 – 04/2026

📁 PROJECTS

Emotion Modulation Framework for LLMs,

NLP, LLM, Transformers, ASR, TTS, Speech AI

- Architected an Emotionally Aligned AI system via prompt engineering, activation steering, and circuit-level interventions, hitting 97.4% emotion accuracy on Mistral-3-3B and 99.4% on Qwen3-4B-Instruct.
- Assembled a multimodal emotion pipeline spanning a RoBERTa emotion classifier (98.8% F1), target classifier (96.6% F1), LoRA-tuned SenseVoice for speech emotion recognition, and emotion-aware TTS via CosyVoice2 and F5-TTS — enabling controllable emotional speech across 6+ categories.

Cold Email Generator - an End-to-End Generative AI

Tool, LangChain, ChromaDB, Flask, Python, RAG, FastAPI

- Designed a full-stack GenAI platform with LLaMA 3.1 and LangChain that automated personalized cold-email outreach, cutting drafting time ~80% per campaign.
- Integrated Retrieval-Augmented Generation (RAG) with a ChromaDB vector store and Flask interface, enabling context-aware emails calibrated to industry, service offerings, and tone — validated across 3 distinct B2B verticals.

Bipedal Robot Locomotion via Reinforcement Learning

- Designed and deployed a hybrid control framework integrating Deep Reinforcement Learning (DRL) algorithms with a low-level Proportional-Derivative (PD) controller to achieve dynamic balance and continuous bipedal walking in PyBullet.
- Evaluated continuous action space actor-critic architectures (SAC, TD3, DDPG), demonstrating that TD3 provided optimal stability, convergence, and gait smoothness across 600,000 training timesteps.
- Formulated multi-objective reward shaping mechanisms balancing forward velocity, posture alignment, and energy efficiency (Cost of Transport) while mitigating fall risks and excessive torque usage.
- Built a high-dimensional observation space processing joint kinematics, base velocities, and foot-contact sensors to handle complex physical dynamics.

🧠 SKILLS

Spoken Languages: German (A2 — Goethe exam completed, result awaited), English (C1), Hindi (Native), Gujarati (Native)

Programming Languages: Python, SQL, Java, C++, C

Technologies: PyTorch, TensorFlow, Scikit-Learn, OpenCV, Stable-Baselines3, Gymnasium, SpaCy, NLTK, Pandas, NumPy, Matplotlib, Git, GitHub, Docker, Linux, Power BI, Tableau, Gazebo Simulation

Concepts: Machine Learning, Deep Learning, Reinforcement Learning, Imitation Learning, Large Language Models, Natural Language Processing, Retrieval Augmented Generation, Generative AI, Agentic AI

🏢 EXPERIENCE

Research Intern, Natural Language and Information Processing Lab, IIT Hyderabad

01/2026 – 05/2026 | Hyderabad, India

PyTorch, Transformers, Llama-3, Mistral-7B, DeepSeek, Context Adaption

- Developed a prompt-driven synthetic data pipeline combining Llama-3 and Mistral-7B to generate 2,000+ culturally localized instruction pairs, reducing manual curation effort by an estimated 60%.
- Benchmarked 5 LLMs (DeepSeek, Qwen, Gemma, Llama, Mistral) across a 4-level word-count prompt taxonomy, quantifying a 23% average drop in instruction-following accuracy at the highest complexity tier — findings directly informed dataset curation strategy.

Research Intern, Centre for Artificial Intelligence and Robotics, IIT Mandi

05/2025 – 08/2025 | Mandi, India

- Developed motion-capture-driven locomotion controllers utilizing Generative Adversarial Imitation Learning (GAIL) coupled with Trust Region Policy Optimization (TRPO), achieving a 15% reduction in trajectory kinematic deviation compared to standard RL baselines across 3 MuJoCo continuous control environments.
- Algorithmic Ablation & Evaluation: Conducted comparative benchmark analysis across model-free continuous control algorithms (SAC, DDPG, and TD3) integrated with customized reward-shaping strategies; demonstrated that TD3 achieved superior gait stability and a 12% improvement in sustained walking distance over SAC.

📄 PUBLICATIONS

Plant Disease Classification with Spectral Signature Taxonomy and Analysis

Software (SSTAS)  , Software Impacts

01/12/2024

Jayswal Hardik A., **Hetvi Desai**, Hasti Vakani, Mithil Mistry, Nilesh Dubey et al. Web of Science - ESCI, Impact Factor: 1.3, Q3.

Symptom-Based Early Detection and Classification of Plant Diseases Using

AI-Driven CNN+KNN Fusion Software (ACKFS)  , Software Impacts

01/12/2024

Jayswal Hardik A., Rishi Sanjaykumar Patel, **Hetvi Desai**, Hasti Vakani, Mithil Mistry, Nilesh Dubey et al. Published in an SCI journal.

Crop Recommendation Systems: Insights, Trends, and Methodological

Approaches  , Springer

22/11/2024

Hetvi Desai, et al. In 9th International Conference on Information and Communication Technology for Competitive Strategies (ICTCS-2024), December 22-23, 2024, Jaipur, India.